

BOOKSTORE – Web Application

A Full Stack MERN Web Application

Submitted by

Name: Vani Sree

Roll No: A24126552125

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Abstract

BookStore is a full-stack online book shopping application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The application enables users to browse books, search by title, filter books by category, view detailed information, add books to a shopping cart, and manage purchases through an interactive interface.

The frontend is built using React.js with React Router and Redux Toolkit for state management. The backend is developed using Express.js and Node.js, while MongoDB is used for storing user and book information. JWT authentication and bcrypt password encryption provide secure user login and registration.

The application offers a responsive user interface that works across desktops, tablets, and mobile devices.

TECHNOLOGIES USED

Frontend

- HTML5
- CSS3
- JavaScript
- React.js
- React Router
- Redux Toolkit

Backend

- Node.js
- Express.js
- MongoDB

Tools

- Visual Studio Code
- Git
- GitHub
- Vercel
- Render

SYSTEM REQUIREMENTS

Hardware

- Processor: Intel i3 or above
- RAM: 4 GB or above
- Storage: 2 GB Free Space

Software

- Windows 10/11
- Node.js
- MongoDB Atlas
- Visual Studio Code
- Git
- Chrome Browser

PROJECT ARCHITECTURE

BookStore

Client (React)

↓

Axios

↓

Express API

↓

MongoDB Atlas

FEATURES

User

- Register
- Login
- Browse Books
- Search Books
- Filter by Category
- View Book Details
- Add to Cart

- Responsive UI

Admin (Backend Ready)

- Add Books
- Update Books
- Delete Books
- View Orders

MODULES

User Authentication

- Registration
 - Login
 - JWT Authentication
 - Password Encryption
-

Book Module

- Display Books
 - Search Books
 - Filter Books
 - Book Details
-

Cart Module

- Add to Cart
 - Remove Items
 - Total Price Calculation
-

Backend Module

- REST APIs
- CRUD Operations
- MongoDB Integration
- Express Routing

DATABASE DESIGN

Users

Field	Type
Name	String
Email	String
Password	String
role	String

Books

Field	Type
Title	String
Author	String
Category	String
Price	Number
Image	String
Description	String
Rating	Number

Orders

Field	Type
User	ObjectId
Books	String
Total Price	Number
Status	String

Reviews

Field	Type
User	ObjectId
Books	ObjectId
Rating	Number
Comment	String

PROJECT STRUCTURE

BookStore

client/

components/

pages/

data/

store/

services/

server/

config/

controllers/

middleware/

models/

routes/

OUTPUT LINK

<https://book-store-blush-iota.vercel.app/>

FUTURE ENHANCEMENTS

- Online Payment Integration
 - Wishlist
 - Order Tracking
 - Admin Dashboard
 - Book Reviews
 - Recommendation System
 - Email Notifications
 - Inventory Management
-

CONCLUSION

The BookStore project demonstrates the practical implementation of the MERN Stack for developing a modern full-stack web application. It integrates a responsive React frontend with an Express and Node.js backend, MongoDB database design, JWT authentication, and RESTful APIs. The project provides a scalable foundation for an online bookstore and showcases essential full-stack development concepts such as CRUD operations, routing, authentication, and state management.